

BAREILLY, IndiaWMO: **421890**Lat: **28.422N**Lon: **79.451E**Elev: **580**StdP: **14.39**Time Zone: **5.50 (IND)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	46.2	48.2	39.7	36.8	60.2	42.5	40.9	59.5	5.6	60.7	5.3	60.5	1.1	270	0.318

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	20.4	105.9	75.0	103.5	75.7	101.1	75.9	84.5	93.1	83.7	92.0	83.1	91.0	3.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.6	173.5	88.4	81.8	169.2	87.8	81.2	165.7	87.3	49.6	94.1	48.7	92.3	47.9	91.3	88.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	7.4	5.6	4.9	40.9	110.3	4.7	1.9	37.5	111.6	34.7	112.8	32.0	113.9	28.6	115.3	
(5)				39.8	86.3	4.8	1.3	36.3	87.2	33.5	88.0	30.9	88.8	27.4	89.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	78.0	58.6	66.5	75.9	86.1	90.0	85.4	85.7	84.5	79.9	71.2	61.6
		DBStd	11.45	4.77	4.64	5.54	4.96	4.28	4.98	3.52	2.93	3.36	3.70	5.05
		HDD50	3	2	0	0	0	0	0	0	0	0	0	1
		HDD65	365	203	33	2	0	0	0	0	0	0	1	126
		CDD50	10221	270	462	802	1084	1241	1099	1106	1035	926	635	360
		CDD65	5106	5	75	339	634	776	751	634	585	461	185	20
		CDH74	64127	18	409	3082	8546	11673	11287	8290	8388	7158	4286	929
		CDH80	34088	0	69	1296	5095	7436	7090	3991	4039	3238	1681	151
(14)	Wind	WSAvg	1.3	0.9	1.5	1.6	1.9	2.0	2.1	1.4	1.1	0.7	0.5	0.7
(15)	Precipitation	PrecAvg	45.5	0.9	1.1	0.5	0.4	0.8	4.7	13.9	12.1	7.7	1.6	0.1
		PrecMax	61.1	3.1	3.9	3.6	2.5	3.2	14.7	22.9	23.2	20.8	8.6	1.1
		PrecMin	24.9	0.1	0.0	0.0	0.0	0.0	1.3	3.0	0.2	0.6	0.0	0.0
		PrecStd	11.4	0.8	1.1	0.8	0.6	1.0	4.4	5.7	5.4	1.6	0.3	0.4
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	76.3	86.4	97.3	105.7	108.0	109.4	99.5	97.1	96.0	93.3	85.8	78.7
		0.4% MCWB	64.0	68.7	73.9	73.9	74.6	76.0	82.5	82.7	80.8	76.8	68.9	64.4
		2% DB	73.2	82.0	93.4	103.3	105.8	106.4	96.7	94.8	94.0	91.3	83.6	75.6
		2% MCWB	62.5	67.0	71.7	73.5	74.6	75.8	82.8	82.6	80.6	75.9	68.6	63.3
		5% DB	70.6	78.9	90.1	100.9	103.4	103.8	94.3	93.2	92.5	89.6	81.8	73.3
		5% MCWB	60.6	65.4	70.3	73.3	74.9	77.8	82.8	82.3	80.4	75.7	67.9	62.6
		10% DB	67.8	76.3	87.2	98.5	101.0	100.7	92.4	91.7	91.0	87.9	79.7	70.8
		10% MCWB	59.0	64.2	69.3	72.6	75.3	78.9	82.5	82.1	80.3	75.1	67.1	61.6
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	66.1	71.0	77.7	83.5	85.1	85.2	85.3	84.7	84.3	81.0	72.6	67.0
		0.4% MCDB	73.7	82.4	92.6	98.9	96.6	95.6	93.1	91.0	91.1	88.2	80.8	75.4
		2% WB	63.8	68.6	74.9	80.7	82.9	83.9	84.3	83.9	82.8	80.0	71.1	65.3
		2% MCDB	71.4	78.9	90.0	96.1	94.7	95.2	92.2	91.0	89.4	87.3	79.6	72.9
		5% WB	61.9	67.0	72.5	77.8	80.9	82.9	83.6	83.4	82.0	78.7	69.8	63.8
		5% MCDB	68.6	76.7	85.4	93.8	94.8	93.7	91.0	90.2	88.7	86.1	78.6	70.8
		10% WB	60.2	65.4	70.7	75.4	79.1	82.0	82.9	82.8	81.4	77.2	68.6	62.2
		10% MCDB	66.1	74.3	83.2	93.3	93.6	92.4	89.8	89.3	88.0	84.8	77.0	68.9
(35)	Mean Daily Temperature Range	MDBR	16.3	19.0	21.0	23.4	20.4	16.4	10.2	10.1	11.5	16.7	18.8	17.2
		5% DB	20.4	21.2	23.2	24.7	23.5	21.7	13.7	12.6	13.7	16.9	19.5	20.1
		MCWBR	11.0	9.4	8.3	8.8	7.5	6.8	4.6	4.4	4.9	6.0	7.7	9.9
		5% WB	18.4	19.6	21.7	22.6	18.6	15.5	11.9	11.0	11.8	13.7	17.8	17.8
(40)	Clear-Sky Solar Irradiance	taub	0.513	0.508	0.504	0.575	0.678	0.735	0.709	0.627	0.583	0.635	0.612	0.557
		taud	1.811	1.880	1.924	1.795	1.630	1.567	1.638	1.826	1.887	1.685	1.670	1.728
		Ebn at Noon	221	239	250	237	214	201	206	222	227	200	190	199
		Edh at Noon	58	59	60	70	82	88	81	67	61	70	67	61
(44)	All-Sky Solar Radiation	RadAvg	866	1329	1780	2066	2129	1817	1438	1476	1505	1428	1131	909
		RadStd	151	109	74	77	107	205	129	157	154	78	68	77

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page